



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!**

Let's make a comparison between C and C#.

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Learning Outcome:

Describe a general-purpose computer programming language, regarding its history and architecture in spoken and written situations.

Ask your partner:

- What do you know about C and C#?
- What programming languages can you name?

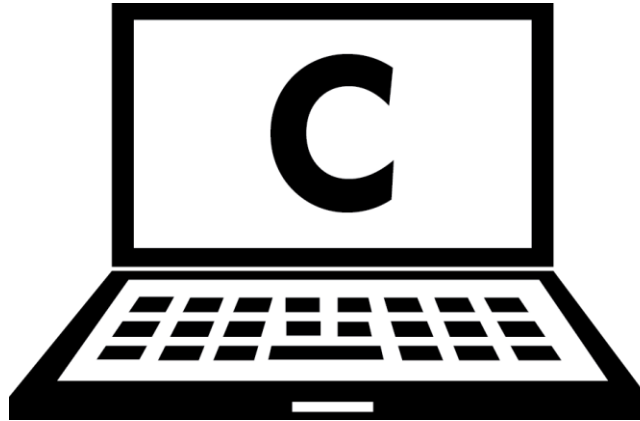
Can Do Statements:

- I can understand written information about the differences between C and C# programming languages.
- I can read ideas about the differences between C and C#.
- I can write information using my own words in a chart to present differences between C and C# programming languages.



Let's Explore!

The teacher will explain the history of C and describe the differences between C and C#.



Instructions: Read the text below.

We have already said that both C and C# are versatile and widely used programming languages. In today's session, we are going to see some characteristics of both of them, and at a point, we will compare them. In any case, we will not say that one is better than the other; they were designed for different purposes. In 1969, a man named Dennis Ritchie designed C language. It came as a structured and standardized programming language mainly used for scientific and management purposes. C organizes files and modules; each file has some specific features like data, processes, and functions. Engineers have used this language for tasks ranging from system programming app to even chip designing or programming for embedded devices.

In 1999, a man named Anders Hejlsberg developed C#. This language is technically a hybrid between C and C++ and is suitable for a wide variety of purposes due to its flexibility. One of its most significant uses is for Windows application development. C# is now almost every developer's first choice for building Windows desktop applications thanks to being user-friendly. It is also widely used for the development of mobile applications, which is one of the fastest growing markets in the programming world. C# can also be used for programming and coding web applications, making it extremely useful for website designers and advertisers alike. When comparing these two languages, we will start by saying that C is a structure-oriented language while C# is an object-oriented one.

Overall, both languages are going to be essential to meet your programming needs. In the programming industry, major companies have preferred to hire people who have the ability to read and use both C and C#. In the future, it is very likely that the way we program will change completely, but these days, C and C# have been and are going to remain two of the most important programming languages.



**Let's Practice!**

Edward identifies the differences among C and C#.

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. It was created to develop operating systems.
2. It is also used for coding web applications
3. It was difficult to write.
4. It was a hybrid from previous languages.
5. Website designers considered it very useful.
6. It was developed first.

C and C++ programming language	C# programming language

**Let's Practice!**

Edward read some information about programming languages.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: was more, solved, had some, was the

1. An updated version of C _____ problems in an easy way.
2. C _____ problems and they were difficult to solve.
3. C# _____ useful for Windows application development.
4. One of C# characteristics _____ flexibility.





Extension Activity

Complete a Venn diagram with the topic seen.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Review the information gathered from this object.
- Complete a Venn diagram.
- In the left and right section, write differences of each C.
- In the middle, write similarities of C, C++ and C#.
- Discuss your Venn diagram with a partner.

Self-Assessment

Yes

No

- I summarize information from this object.
- I make comparisons by using a Venn diagram.
- I discuss my findings with a partner.

☐	☐
☐	☐
☐	☐



Target Words

- **standardized** (adjective): adopt as an standard
e.g. It came as a structured and standardized programming language mainly used for scientific and management purposes.
- **function** (noun): a code with a specific property
e.g. C organizes files and modules; each file has some specific features like data, processes, and functions.
- **hybrid** (noun): something made combining two different elements
e.g. In 1999, a man named Anders Hejlsberg developed C#. This language is technically a hybrid between C and C++ and is suitable for a wide variety of purposes due to its flexibility.
- **flexibility** (noun): versatility
e.g. In 1999, a man named Anders Hejlsberg developed C#. This language is technically a hybrid between C and C++ and is suitable for a huge variety of purposes due to its flexibility.
- **range** (verb): to go from one to another limit
e.g. Engineers have used this language for tasks ranging from system programming app to even chip designing or programming for embedded devices.



Learning Points

Present Perfect Versus Past Simple

The present perfect is used to talk about unfinished actions that started in the past. The past simple is used to talk about actions that finished in the past.

We use the **present perfect** to talk about actions that started in the past and still occur or have connection with the present. We use the **past simple** for actions that started and finished in the past.

We also use the **present perfect** to talk about finished actions in the past that have a result in the present.

Present perfect:

Interrogative form:

Have/has + subject + **verb past participle** + complement.

Example: **Have** you **used** C# lately?

Affirmative form:

Subject + **have/has** + **verb past participle** + complement.

Example: I **have started** a new project with C#.

Negative form:

Subject + **have/has** + not + **verb past participle** + complement.

Example: I **have not started** the new project yet!

Simple past:

Interrogative form:

Did + subject + **verb base form** + complement.

Example: **Did** you **use** C# in your last project?

Affirmative form:

Subject + **verb past simple** + complement.

Example: I **started** a new project with C#.

Negative form:

Subject + **did** + not + **verb base form** + complement.

Example: I **did not start** the project yesterday!

Examples

- *We have already said that both C and C# are versatile and widely used programming languages.*
- *Engineers have used this language for tasks ranging from system programming app to even chip designing or programming for embedded devices.*
- *In the programming industry, major companies have preferred to hire people who have the ability to read and use both C and C#.*





Learning Points

Future With Will and Going To

You can use **will** and **going to** to talk about future. However, there are some slight differences that make you choose one or the other.

We use the future simple with **will** for:

- Predictions without evidence
- Decisions made at the moment of speaking
- Promises or requests

Affirmative form:

Subject **will** + **verb base form** + complement.

Example: *I **will** start a new project with C#.*

Negative form:

Subject + **will** + not + **verb base form** + complement.

Example: *I **will not use** C++ for this project.*

Interrogative form:

Will + subject + **verb base form** + complement.

Example: ***Will** you **use** C# in your next project?*

We use the future with **going to** for:

- Something you had decided to do before
- A prediction made on evidence (that we see or hear)

Affirmative form:

Subject + verb to be **going to** + **verb base form** + complement.

Example: *I **am going to use** C++ for all my projects this year.*

Negative form:

Subject + + verb to be + not + **going to** + **verb base form** + complement.

Example: *I **am not going to program** this task today; I'll do it tomorrow.*

Interrogative form:

Verb to be + subject + **going to** + **verb base form** + complement.

Example: *Are you **going to use** C# in your next project?*

Examples

- *In today's session, we are going to see some characteristics of both of them, and at a point, we will compare them.*
- *Overall, both languages are going to be essential to meet your programming needs.*
- *We will start by saying that C is a structure-oriented language while C# is an object-oriented one.*

